

Chapter 23. Business Acumen, KPIs & Metrics

How a business makes money, the metrics every function watches, and how to diagnose a number instead of guessing at it.

Approved chapter v1

Approved 18 September 2026

Real revenue and delivery data; modelled financial statements clearly labelled

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Chapter 23. Business Acumen, KPIs & Metrics

Part II — The Analyst

Chapter at a glance

You will learn to: trace how a business turns cash into stock into sales and back into cash · read a P&L, a balance sheet, and a cash flow statement well enough to hold a conversation with Finance · explain why a profitable company can still run out of money · use the core metrics of sales, marketing, finance, operations, and customer teams, and compute each one correctly · build a KPI tree that decomposes revenue into numbers a team can actually act on · diagnose a revenue change with a disciplined walk down that tree instead of a guess · tell a good metric from a vanity metric, and see how targets corrupt measures · write a metric definition precise enough that two teams get the same number.

Before you start: Chapter 4 (percentages and the arithmetic of business), Chapter 18 (pandas), Chapter 21 (distributions and averages), and Chapter 22 (whether a change is real). Nothing here assumes an accounting or finance background.

Time needed: 15–18 hours, spread over two weeks.

Tools: Python 3.13 or 3.14 with `pandas` (run here on Python 3.12, `pandas 3.0.2`), or Excel. A calculator and a willingness to read a financial statement slowly.

Practice data: the full Riverstone dataset, Chapter 21's delivery-time data, and `companion/ch23/ : financials_fy2025.md` (Riverstone's FY2025 profit & loss, balance sheet, and cash flow statement), `monthly_revenue_2025.csv`, and `marketing_2025.csv`. Revenue, cost of goods sold, gross margin, retention, and every operational number in this chapter come directly from the real order and delivery data and match every earlier chapter exactly. Everything below gross profit on the P&L, the whole balance sheet, the cash flow statement, and the marketing and NPS figures are invented for this chapter, because Riverstone's ERP data has no general ledger, no balance sheet, and no marketing system. Each table says which is which.

Why this matters

Every chapter so far has taught you to get a number right. This chapter teaches you to know which numbers matter, and why.

An analyst who can write a flawless query but doesn't know that a sale isn't cash until it's collected will build a beautiful chart that misses the point of the meeting. An analyst who reports "conversion is up 12%" without asking up against what, over what period, and whether anyone changed the

definition halfway through, will get the number believed and the decision wrong. Business acumen is not a soft skill bolted onto the technical ones; it's the thing that tells you which query to write in the first place.

This chapter gives you four things: how to read the three financial statements that describe any company, the standard metrics of the five functions you'll work with most, a method for building and reading a KPI tree, and the judgment to know when a metric is helping and when it's being gamed. Riverstone is used throughout, with its real 2025 sales data and an invented but realistic set of financial statements, so every ratio in this chapter is something you can recompute yourself.

In plain English

Imagine you take over your family's shop for a month while everyone is away. Three things worry you immediately, and they are the three financial statements in disguise.

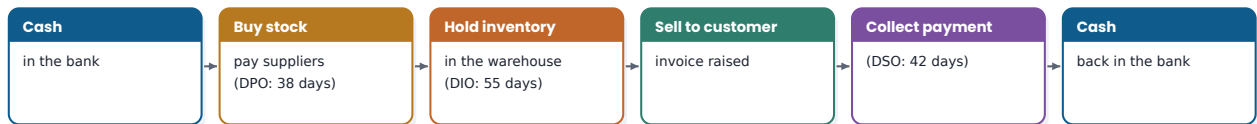
- **"Am I making money?"** You add up what came in from customers and take away what you spent on stock, rent, and the assistant's wages. That's the **profit & loss statement**: a video of the month.
- **"What do we own, and what do we owe?"** You check the stockroom, the cash box, and the ledger of who owes you and who you owe. That's the **balance sheet**: a photograph taken at the end of the month.
- **"Do I have enough cash to pay the supplier on Friday?"** You might be profitable on paper and still short of cash, if customers haven't paid you yet and the supplier wants payment now. That's the **cash flow statement**, and it's the one that keeps shopkeepers awake at night.

A **metric** is a number you've decided to watch, the way the shopkeeper watches "cash in the box" and "stock on the shelf". A good metric tells you something you can act on. A **KPI tree** is writing down that if takings are low, it's because either fewer customers walked in, or each one bought less — which is exactly how you'd think it through standing behind the counter, just written down so a team can each own one branch.

23.1 How a business makes money: the operating cycle

Every trading business, however it's dressed up, runs the same cycle: it turns cash into stock, stock into a sale, and a sale back into cash. Analysts who understand this cycle ask better questions than analysts who only understand the tables.

The operating cycle: how cash becomes stock becomes cash again



Cash conversion cycle = DIO + DSO – DPO = 55 + 42 – 38 = 59 days

Riverstone's cash is tied up in stock and unpaid invoices for 59 days before it becomes cash again — even though the company is profitable. A profitable company can still run out of cash if this cycle is long and growing fast.

Figure 23.1 – The operating cycle. Riverstone's cash is committed for 59 days between paying a supplier and collecting from a customer, even though the company is profitable.

- **DIO (Days Inventory Outstanding):** how long stock sits in the warehouse before it sells. Riverstone: 55 days.
- **DSO (Days Sales Outstanding):** how long after a sale it takes to collect the cash. Riverstone: 42 days.
- **DPO (Days Payable Outstanding):** how long Riverstone takes to pay its own suppliers. Riverstone: 38 days.
- **Cash conversion cycle (CCC) = DIO + DSO – DPO.** Riverstone's is **59 days**: cash goes out to buy stock, and doesn't fully come back for two months.

A business that's growing fast needs *more* cash tied up in this cycle every month, not less, because next month's stock and receivables are bigger than this month's. That's why "we're profitable but we need a loan" is a completely ordinary sentence, and why an analyst who only looks at the P&L will misread it as a contradiction.

23.2 The profit & loss statement

The **P&L** (also income statement) covers a period — a month, a quarter, a year — and answers "did we make money, and on what?"

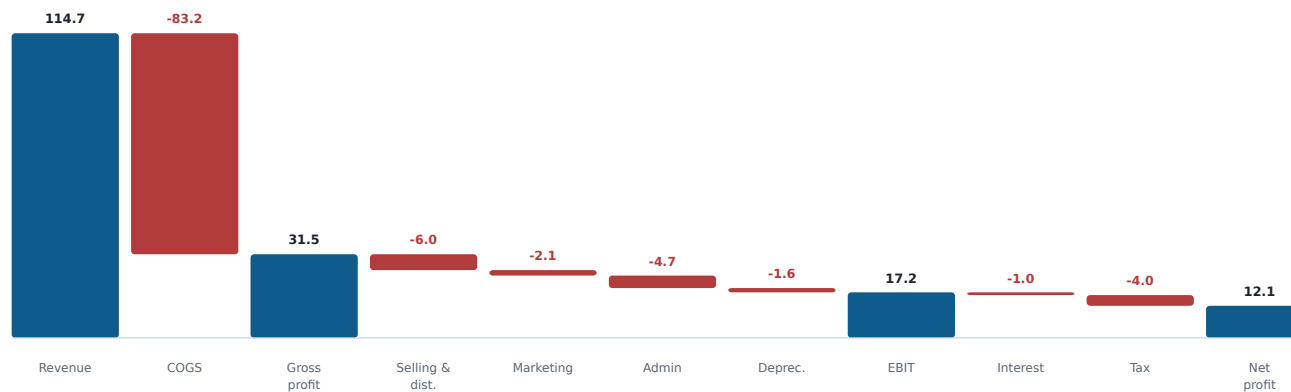
```
import pandas as pd
pd.set_option("display.width", 100)

REVENUE, COGS = 1_146_641_651, 831_803_000
gross_profit = REVENUE - COGS
selling_dist, marketing, admin, depreciation = 59_625_366, 20_639_550, 47_012_308, 16_052_983
opex = selling_dist + marketing + admin + depreciation
ebit = gross_profit - opex
interest = 10_319_775
pbt = ebit - interest
tax = round(pbt * 0.25)
pat = pbt - tax

for label, value in [("Revenue", REVENUE), ("COGS", -COGS), ("Gross profit", gross_profit),
                    ("Operating expenses", -opex), ("EBIT", ebit), ("Interest", -interest),
                    ("Profit before tax", pbt), ("Tax", -tax), ("Profit after tax", pat)]:
    print(f"{label:<20} ₹{value:>15,}    ({value/REVENUE*100:>5.1f}% of revenue)")
```

Revenue	₹ 1,146,641,651	(100.0% of revenue)
COGS	₹ -831,803,000	(-72.5% of revenue)
Gross profit	₹ 314,838,651	(27.5% of revenue)
Operating expenses	₹ -143,330,207	(-12.5% of revenue)
EBIT	₹ 171,508,444	(15.0% of revenue)
Interest	₹ -10,319,775	(-0.9% of revenue)
Profit before tax	₹ 161,188,669	(14.1% of revenue)
Tax	₹ -40,297,167	(-3.5% of revenue)
Profit after tax	₹ 120,891,502	(10.5% of revenue)

Riverstone's FY2025 P&L, top to bottom (₹ crore)



Gross margin 27.5% · EBIT margin 15.0% · net margin 10.5%. Cost of goods sold is real (the order data); everything else is modelled for this chapter.

Figure 23.2 – Riverstone's FY2025 P&L. Revenue and COGS are real; everything below gross profit is modelled for this chapter, sized to a plausible mid-size distributor.

Line	What it means	Riverstone FY2025
Revenue (net sales)	What customers paid, after discounts and returns	₹114.66 crore
COGS (cost of goods sold)	What the goods themselves cost	₹83.18 crore
Gross profit / margin	Revenue minus COGS; the margin on the product itself	₹31.48 crore, 27.5%
Operating expenses (opex)	Selling, marketing, admin, depreciation: running the business	₹14.33 crore
EBIT (operating profit)	Profit from operations, before financing and tax	₹17.15 crore, 15.0%
Interest	Cost of borrowed money	₹1.03 crore
Tax	Statutory tax on profit	₹4.03 crore
PAT (net profit, "the bottom line")	What's left for the owners	₹12.09 crore, 10.5%

Three distinctions worth knowing cold:

- **Gross margin versus net margin.** Gross margin (27.5%) is about the product; net margin (10.5%) is about the whole business. A retailer and a software company can have wildly different gross margins and similar net margins, because they spend differently below the line.

- **EBIT versus EBITDA.** EBITDA adds depreciation and amortisation back to EBIT, which strips out the effect of how assets are financed and aged. It's popular for comparing companies with different asset bases; it is not cash profit, and treating it as such is how highly leveraged, unprofitable companies get described as "healthy".
- **Revenue is not cash.** A sale is booked when it's made, not when it's paid for (this is **accrual accounting**, the standard for anyone but the smallest business). That single fact is why the next two sections exist.

Watch out: percentages of revenue hide absolute size. "Marketing is 1.8% of revenue" sounds small; it is ₹2.06 crore, hired-analysts'-salaries money. Always look at both the percentage and the rupee figure before deciding something is immaterial.

23.3 The balance sheet

The **balance sheet** is a snapshot at a single date: what the company owns (**assets**), what it owes (**liabilities**), and the difference, which belongs to the owners (**equity**). It always balances, by definition: **Assets = Liabilities + Equity**.

```
cash, receivables, inventory, other_ca = 69_945_141, 131_942_327, 125_340_178, 17_199_625
current_assets = cash + receivables + inventory + other_ca
ppe, other_nca = 355_458_912, 22_932_833
total_assets = current_assets + ppe + other_nca

payables, st_debt, other_cl = 86_598_668, 97_464_540, 34_399_250
current_liabilities = payables + st_debt + other_cl
lt_debt = 126_130_582
total_liabilities = current_liabilities + lt_debt
equity = total_assets - total_liabilities

print(f"total assets      ₹{total_assets:,}")
print(f"total liabilities ₹{total_liabilities:,}")
print(f"equity             ₹{equity:,}")
print(f"check: liabilities + equity = ₹{total_liabilities + equity:,} (must equal total assets)")
print(f"\nworking capital = current assets - current liabilities = ₹{current_assets - current_liabilities:,}")
print(f"current ratio    = current assets ÷ current liabilities = {current_assets/current_liabilities:.2f}")
print(f"debt-to-equity    = total liabilities ÷ equity = {total_liabilities/equity:.2f}")
```

```
total assets      ₹722,819,016
total liabilities ₹344,593,040
equity            ₹378,225,976
check: liabilities + equity = ₹722,819,016 (must equal total assets)

working capital = current assets - current liabilities = ₹125,964,813
current ratio   = current assets ÷ current liabilities = 1.58
debt-to-equity  = total liabilities ÷ equity = 0.91
```

Term	Meaning	Riverstone
Current assets	Cash, or convertible to cash within a year: cash, receivables, inventory	₹34.44 crore

Term	Meaning	Riverstone
Current liabilities	Due within a year: payables, short-term debt	₹21.85 crore
Working capital	Current assets – current liabilities: the cushion for day-to-day operations	₹12.60 crore
Current ratio	Current assets ÷ current liabilities: can short-term obligations be met?	1.58
Debt-to-equity	Total liabilities ÷ equity: how much of the company is borrowed	0.91

Why an analyst should care: **receivables and inventory are analyst territory**. Both sit on the balance sheet, both are driven by the operational decisions Chapters 13 and 21 analyze (who gets credit terms, how much stock to hold, how fast orders ship), and both are exactly where a growth story turns into a cash problem. If receivables are growing faster than revenue, customers are being allowed more time to pay, or some of them aren't paying at all (Chapter 14's territory, applied to money owed rather than money already booked).

23.4 Cash flow: why profit isn't cash

The **cash flow statement** reconciles profit (an accounting figure) with the actual change in the cash balance, in three sections.

```
pat, depreciation = 120_891_502, 16_052_983
receivables_change, inventory_change, payables_change = 4_586_567, 6_879_850, 3_439_925
cfo = pat + depreciation - receivables_change - inventory_change + payables_change

capex = 51_598_874
cfi = -capex

debt_drawn, dividend, loan_repayment = 22_932_833, 42_312_026, 9_173_133
cff = debt_drawn - dividend - loan_repayment

net_change = cfo + cfi + cff
print(f"Cash flow from operations (CFO): ₹{cfo:,}")
print(f"Cash flow from investing (CFI): ₹{cfi:,}")
print(f"Cash flow from financing (CFF): ₹{cff:,}")
print(f"Net change in cash: ₹{net_change:,}")
print(f"\nProfit after tax was ₹{pat:,}, but cash rose by only ₹{net_change:,}.")
print(f"The difference is mostly capital spending (₹{capex:,}) and the dividend (₹{dividend:,}).")
```

```
Cash flow from operations (CFO): ₹128,917,993
Cash flow from investing (CFI): ₹-51,598,874
Cash flow from financing (CFF): ₹-28,552,326
Net change in cash: ₹48,766,793
```

```
Profit after tax was ₹120,891,502, but cash rose by only ₹48,766,793.
The difference is mostly capital spending (₹51,598,874) and the dividend (₹42,312,026).
```

- **CFO (operations):** profit, adjusted for non-cash items (add back depreciation) and for money tied up in working capital (growing receivables and inventory use cash; growing payables frees it). Riverstone's CFO of ₹12.89 crore is healthily above its PAT of ₹12.09 crore.
- **CFI (investing):** money spent on or raised from long-term assets — warehouses, vehicles, equipment. Almost always negative for a growing company, and that's fine.

- **CFF (financing):** money from or to lenders and owners — loans drawn or repaid, dividends paid.

The single most important habit this section can give you: when someone says "we made ₹12 crore profit", ask "and how much of that is in the bank?" A company can report a profit and still fail, if that profit is sitting in unpaid invoices and a growing stockroom. A company can report a loss and still be safe, if it's investing heavily in growth that hasn't billed yet. Profit is an opinion; cash is a fact, delayed by exactly the operating cycle in section 23.1.

23.5 Sales metrics

Sales metrics describe the **pipeline**: the journey from a lead to a paying customer. Riverstone's CRM has real (if small) pipeline data for 2025.

```
import pandas as pd
leads = pd.read_parquet("../full/leads.parquet")
hist = pd.read_parquet("../full/lead_stage_history.parquet")

reached = {stage: hist.groupby("lead_id")["stage"].apply(lambda s: stage in set(s)).sum()
            for stage in ["New", "Contacted", "Quoted", "Won", "Lost"]}
print("funnel:", reached)

won, lost = reached["Won"], reached["Lost"]
print(f"\nwin rate (won / (won + lost))      = {won/(won+lost)*100:.1f}%")
print(f"win rate (won / total leads)         = {won/reached['New']*100:.1f}%")

cycle = (hist[hist.stage == "Won"][["lead_id", "entered_at"]].rename(columns={"entered_at":
"won_at"}))
        .merge(hist[hist.stage == "New"][["lead_id", "entered_at"]].rename(columns={"entered_at":
"created"}),
              on="lead_id")
cycle["days"] = (cycle.won_at - cycle.created).dt.days
print(f"average sales cycle                 = {cycle.days.mean():.1f} days")
print(leads.source.value_counts())
```

```
funnel: {'New': np.int64(43), 'Contacted': np.int64(22), 'Quoted': np.int64(14), 'Won':
np.int64(6), 'Lost': np.int64(4)}

win rate (won / (won + lost))      = 60.0%
win rate (won / total leads)      = 14.0%
average sales cycle                = 25.5 days
source
Website                27
Referral                5
Trade fair             5
Cold call              3
IndiaMART listing      3
Name: count, dtype: int64
```

Metric	Formula	Riverstone 2025	Watch for
Pipeline value	Sum of open deals' expected value	— (not tracked yet)	Stale deals inflate it; age the pipeline
Win rate	Won ÷ (Won + Lost)	60%	Won ÷ <i>all</i> leads (14%) is a different, also-useful number — say which one

Metric	Formula	Riverstone 2025	Watch for
Sales cycle	Days from first contact to close	25.5 days	Skewed by a few very slow deals; report the median too
Conversion by stage	Share reaching each stage	New 100% → Contacted 51% → Quoted 33% → Won 14%	The stage with the biggest drop is where to invest
Average order value (AOV)	Revenue ÷ orders	₹24,735.56 (Chapter 16)	Mean is pulled up by large Wholesale orders (Chapter 21)
Quota attainment	Actual ÷ target, per rep or team	98.3% company-wide (Chapter 16)	An average hides reps far above and far below target

Two-thirds of Riverstone's 43 leads came through the website, which is a finding in itself: a small, informal pipeline where most volume is inbound, and where the "sales cycle" for a self-serve reorder from an existing account (which never touches this CRM at all) is a completely different, much shorter process. **A metric only describes what it measures**, and most of Riverstone's revenue comes from repeat orders that never enter a pipeline stage.

23.6 Marketing metrics

```
mk = pd.read_csv("marketing_2025.csv")
print(mk[["month", "marketing_spend", "new_customers", "cac", "roas"]])
print(f"\ntotal spend      ₹{mk.marketing_spend.sum():,}")
print(f"new customers      {mk.new_customers.sum():,}")
print(f"blended CAC          ₹{mk.marketing_spend.sum()/mk.new_customers.sum():.0f}")
print(f"average ROAS        {mk.attributed_revenue.sum()/mk.marketing_spend.sum():.1f}x")
```

	month	marketing_spend	new_customers	cac	roas
0	2025-01	518896	49	10590.0	4.36
1	2025-02	484256	53	9137.0	5.56
2	2025-03	525922	75	7012.0	7.37
3	2025-04	554956	65	8538.0	5.52
4	2025-05	522505	73	7158.0	5.35
5	2025-06	377585	38	9936.0	4.68
6	2025-07	341529	46	7425.0	5.18
7	2025-08	485093	64	7580.0	5.80
8	2025-09	661445	76	8703.0	5.46
9	2025-10	693425	83	8355.0	5.45
10	2025-11	669940	90	7444.0	6.35
11	2025-12	449210	55	8167.0	6.11
total spend		₹6,284,762			
new customers		767			
blended CAC		₹8,194			
average ROAS		5.6x			

Metric	Formula	Riverstone 2025	Watch for
The funnel	Impressions → clicks → leads → customers	5,650 leads → 767 new customers	Report the rate <i>between</i> each stage, not just the ends

Metric	Formula	Riverstone 2025	Watch for
CAC (customer acquisition cost)	Marketing spend ÷ new customers	₹8,194	Excludes sales salaries and time; a "fully loaded" CAC includes them and is always higher
ROAS (return on ad spend)	Attributed revenue ÷ spend	5.6x , monthly range 4.4–7.4x	Revenue attributed to marketing is a judgment call, not a fact (next paragraph)
Payback period	CAC ÷ monthly gross profit per customer	See section 23.9	Long paybacks strain cash even when the unit economics work

Attribution is the honest weak point of every marketing metric. When a customer sees an ad, visits the website, gets a call from a rep, and orders three months later, which channel "caused" the sale? Common models — **last-touch** (credit the final interaction), **first-touch** (credit the first), and **multi-touch** (split credit across all of them) — can each tell a defensibly different story from the same data, and none is simply correct in an absolute sense. State the model whenever you report CAC or ROAS, the way you'd state units.

Watch out: CAC and LTV need the same population. CAC is measured on the customers marketing actually brought in *this year*. Comparing it against the lifetime value of the whole customer base — which includes decades-old accounts marketing never touched — silently compares two different populations. Section 23.9 shows the fix.

23.7 Finance metrics

Some of these appeared already; here they're gathered as the vocabulary Finance will use with you.

```
gross_margin = 314_838_651 / 1_146_641_651
contribution_margin_pct = 0.31          # illustrative: gross margin plus variable selling costs
ccc = 55 + 42 - 38
```

```
print(f"gross margin          {gross_margin*100:.1f}%")
print(f"EBIT margin          {171_508_444/1_146_641_651*100:.1f}%")
print(f"net margin           {120_891_502/1_146_641_651*100:.1f}%")
print(f"cash conversion cycle {ccc} days")
print(f"return on assets      {120_891_502/722_819_016*100:.1f}%")
print(f"return on equity      {120_891_502/378_225_976*100:.1f}%")
```

```
gross margin      27.5%
EBIT margin       15.0%
net margin        10.5%
cash conversion cycle 59 days
return on assets  16.7%
return on equity   32.0%
```

Metric	Formula	Meaning
Gross margin	Gross profit ÷ revenue	How much of each rupee survives the cost of the product

Metric	Formula	Meaning
Contribution margin	(Revenue – all variable costs) ÷ revenue	What's left to cover fixed costs and profit, per unit sold
EBIT / net margin	Explained in section 23.2	Operating and overall profitability
DSO / DPO / DIO / CCC	Section 23.1	How long cash is tied up
Return on assets (ROA)	Net profit ÷ total assets	How efficiently assets generate profit
Return on equity (ROE)	Net profit ÷ equity	The return the owners are earning on their stake

Fixed versus variable costs is worth a moment. A **fixed cost** (rent, most salaries) doesn't change with volume in the short run; a **variable cost** (materials, packaging, commission) scales with each unit. Contribution margin, not gross margin, is what tells you whether taking one more order is worth it, because gross margin can include fixed costs (like factory overhead allocated per unit) that don't actually change if you make one more crate.

23.8 Operations metrics

Chapter 21's delivery data is Riverstone's real operational record, and it already contains the standard logistics metrics.

```
deliveries = pd.read_csv("../ch21/delivery_times_2025.csv")

otif = deliveries["on_time"].mean()          # "in full" is assumed here: no partial-shipment
data
print(f"On-Time In Full (OTIF), proxy      : {otif*100:.1f}%")
print(deliveries.groupby("branch")["on_time"].mean().mul(100).round(1))
print(f"\nmedian delivery time           : {deliveries['delivery_days'].median():.1f} days")
print(f"orders delivered                  : {len(deliveries):,}")
```

```
On-Time In Full (OTIF), proxy      : 81.8%
branch
Bengaluru      78.0
Delhi          80.0
Kolkata        68.1
Mumbai H0      90.8
Name: on_time, dtype: float64

median delivery time           : 3.8 days
orders delivered               : 45,040
```

Metric	Formula	Riverstone 2025	Notes
OTIF (on-time, in-full)	Orders delivered on time and complete ÷ total	81.8% (on-time only; "in full" isn't tracked)	The two failure modes are different problems; measure both if you can
Inventory turns	COGS ÷ average inventory	₹83.18 cr ÷ ₹12.53 cr ≈ 6.6 times a year	Too high risks stockouts; too low ties up cash (section 23.1)

Metric	Formula	Riverstone 2025	Notes
Fill rate	Units shipped ÷ units ordered	— (not tracked)	Distinct from OTIF: a full but late order fails OTIF, not fill rate
OEE (Overall Equipment Effectiveness)	Availability × performance × quality	— (a plant metric; not applicable to a distributor)	Common in manufacturing; each of the three factors is itself worth decomposing

Kolkata's 68.1% OTIF against Mumbai HO's 90.8% is the same finding as Chapter 21, restated in the vocabulary operations teams use in every industry. That repetition is the point: **the underlying skill — profile the distribution, don't just report the average — is the same whether you call the measure "delivery time" or "OTIF".**

23.9 Customer metrics

```
customers_2024 = set(pd.read_parquet("../full/order_items.parquet").merge(
    pd.read_parquet("../full/orders.parquet"), on="order_id"
).query("order_date >= '2024-01-01' and order_date < '2025-01-01' and status != 'Cancelled'")
["customer_id"])
lines =
pd.read_parquet("../full/order_items.parquet").merge(pd.read_parquet("../full/orders.parquet"),
on="order_id")
lines["net_revenue"] = lines.quantity * lines.unit_price * (1 - lines.discount_pct / 100)
sales_2025 = lines[(lines.order_date >= "2025-01-01") & (lines.status != "Cancelled")]
customers_2025 = set(sales_2025["customer_id"])

retained = customers_2024 & customers_2025
churned = customers_2024 - customers_2025
retention = len(retained) / len(customers_2024)

print(f"active 2024: {len(customers_2024):,}   active 2025: {len(customers_2025):,}")
print(f"retained: {len(retained):,}   churned: {len(churned):,}")
print(f"retention rate: {retention*100:.1f}%   churn rate: {(1-retention)*100:.1f}%")

new_customer_revenue =
sales_2025[~sales_2025.customer_id.isin(customers_2024)].groupby("customer_id")
["net_revenue"].sum().mean()
gross_margin, horizon_years, cac = 0.275, 5, 8194
ltv = new_customer_revenue * gross_margin * horizon_years
print(f"\naverage first-year revenue, new customers: ₹{new_customer_revenue:,.0f}")
print(f"LTV (5-year horizon, {gross_margin*100:.0f}% margin): ₹{ltv:,.0f}")
print(f"CAC: ₹{cac:,.0f}   LTV:CAC = {ltv/cac:.1f} : 1")
```

```
active 2024: 4,104   active 2025: 4,599
retained: 3,827   churned: 277
retention rate: 93.3%   churn rate: 6.7%
```

```
average first-year revenue, new customers: ₹157,617
LTV (5-year horizon, 28% margin): ₹216,723
CAC: ₹8,194   LTV:CAC = 26.4 : 1
```

Metric	Formula	Riverstone 2025	Watch for
Retention rate	Customers active in both periods ÷ active in the first	93.3%	High for B2B distribution: switching suppliers has real friction
Churn rate	1 – retention	6.7%	Small percentages compound: 6.7% a year is a third of the base gone in five years
LTV (customer lifetime value)	Annual revenue × gross margin × customer lifetime	₹2.17 lakh (5-year horizon)	Never use the uncapped 1/churn lifetime (14.8 years here) — see below
LTV : CAC	LTV ÷ CAC	26.4 : 1	A ratio above roughly 3:1 is usually called healthy; this one is unusually high (next paragraph)
NPS (Net Promoter Score)	% promoters (score 9–10) – % detractors (0–6)	+34 (quarterly survey)	A single number from a ten-point question; see the caution below

Watch out: the naive LTV formula overclaims. "LTV = annual value × margin × (1 ÷ churn rate)" gives Riverstone a 14.8-year customer lifetime and an LTV of ₹6.4 lakh — nobody should plan around fourteen years of an unproven assumption holding. **Cap the horizon** (three to five years is standard) and say so. The 5-year-capped LTV here (₹2.17 lakh) is a third of the naive figure, and it's the one to defend in a meeting.

Riverstone's LTV:CAC of 26:1 is far above the textbook "3:1 is healthy" benchmark, and a good analyst treats a suspiciously good number the same way as a suspiciously bad one (Chapter 22): check it before repeating it. Here it holds up, and it has a real explanation — B2B distribution relationships are sticky, switching costs are high, and a customer's *first* order is small relative to what they buy once trust is established — but it also suggests Riverstone could profitably spend far more on acquiring customers than it currently does. Not every extreme number is an error; some are the actual finding.

NPS's blind spot: it collapses ten points of opinion, from many different customers, into one number. A promoter (9–10) and a detractor (0–6) each disappear into a bucket that says nothing about *why*, and identical NPS scores can hide completely different underlying distributions (a company with everyone at 7 versus a company split evenly between 10s and 3s can score similarly). Use it as a trend, watch the *comments* that come with it, and never let it stand alone as "customer health".

23.10 KPI trees and the North Star

A **KPI tree** decomposes one headline number into the factors that multiply or add together to produce it, until each branch is small enough that one team can own it.

A KPI tree: revenue decomposed into things a team can act on

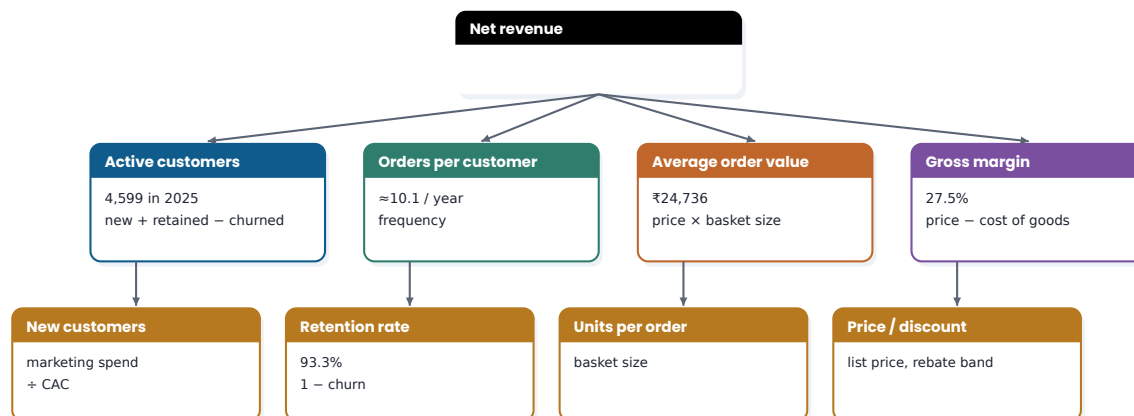


Figure 23.3 – Riverstone's revenue tree. Each first-level branch is owned by a different function; the second level is where a diagnosis (section 23.11) usually lands.

```

active_customers = 4599
orders_per_customer = 46356 / active_customers
aov = 24735.56
reconstructed = active_customers * orders_per_customer * aov
print(f"active customers      {active_customers:,}")
print(f"orders per customer      {orders_per_customer:.2f}")
print(f"average order value        ₹{aov:,.2f}")
print(f"customers × freq × AOV      ₹{reconstructed:,.2f}")
print(f"actual 2025 revenue        ₹1,146,641,651.25")
  
```

active customers	4,599
orders per customer	10.08
average order value	₹24,735.56
customers × freq × AOV	₹1,146,641,619.36
actual 2025 revenue	₹1,146,641,651.25

The identity **Revenue = Active customers × Orders per customer × Average order value** holds exactly, which is what makes it useful: every rupee of revenue change must show up as a change in one or more of those three factors (or in gross margin, if you're decomposing profit rather than revenue). Each of those, in turn, decomposes further: active customers is new customers plus retained customers minus churned ones; average order value is units per order times price, less any discount.

Choosing a North Star metric. Some companies also name one **North Star metric**: a single number, upstream of revenue, that captures whether the business is delivering value (a marketplace might track completed transactions; Riverstone might track active customers, since a customer who orders at all tends to become a repeat one). A North Star is a communication tool, not a replacement for the tree — it's the one branch senior leadership watches weekly, while the rest of the tree stays with the teams that own it.

Building one for your own business, in order:

1. Write the headline number as a **product or sum** of measurable factors — not a vague list, an actual formula that reconstructs the total.
2. Check the identity holds on real data, as above. If it doesn't, a factor is missing or miscounted.

3. Decompose each factor one more level, stopping when a branch is something one team can move without needing another team's help.
4. Name an owner for each branch. A branch with no owner is a fact, not a KPI.

23.11 Diagnosing a change

Riverstone's revenue fell from ₹8.72 crore in May 2025 to ₹5.20 crore in June — a genuine, well-documented seasonal trough (Chapters 11 and 15). Rather than reporting "revenue fell 40%", walk it down the tree.

```
may = {"customers": 2885, "orders": 3658, "revenue": 87_249_568}
jun = {"customers": 2723, "orders": 3240, "revenue": 51_972_483}
for d in (may, jun):
    d["opc"] = d["orders"] / d["customers"]
    d["aov"] = d["revenue"] / d["orders"]

customer_effect = (jun["customers"] - may["customers"]) * may["opc"] * may["aov"]
opc_effect      = jun["customers"] * (jun["opc"] - may["opc"]) * may["aov"]
aov_effect      = jun["customers"] * jun["opc"] * (jun["aov"] - may["aov"])
total_change = jun["revenue"] - may["revenue"]

print(f"May: {may['customers']:,} customers × {may['opc']:.3f} orders/cust × ₹{may['aov']:,} AOV = ₹{may['revenue']:,}")
print(f"June: {jun['customers']:,} customers × {jun['opc']:.3f} orders/cust × ₹{jun['aov']:,} AOV = ₹{jun['revenue']:,}")
print(f"\nEffect of fewer customers: ₹{customer_effect:,.0f}")
print(f"Effect of fewer orders/customer: ₹{opc_effect:,.0f}")
print(f"Effect of lower average order: ₹{aov_effect:,.0f}")
print(f"Sum of effects: ₹{customer_effect+opc_effect+aov_effect:,.0f}")
print(f"Actual change: ₹{total_change:,.0f}")
```

```
May: 2,885 customers × 1.268 orders/cust × ₹23,851.71 AOV = ₹87,249,568
June: 2,723 customers × 1.190 orders/cust × ₹16,040.89 AOV = ₹51,972,483

Effect of fewer customers: ₹-4,899,283
Effect of fewer orders/customer: ₹-5,070,734
Effect of lower average order: ₹-25,307,069
Sum of effects: ₹-35,277,085
Actual change: ₹-35,277,085
```

Diagnosing the May → June dip: a chain-linked bridge

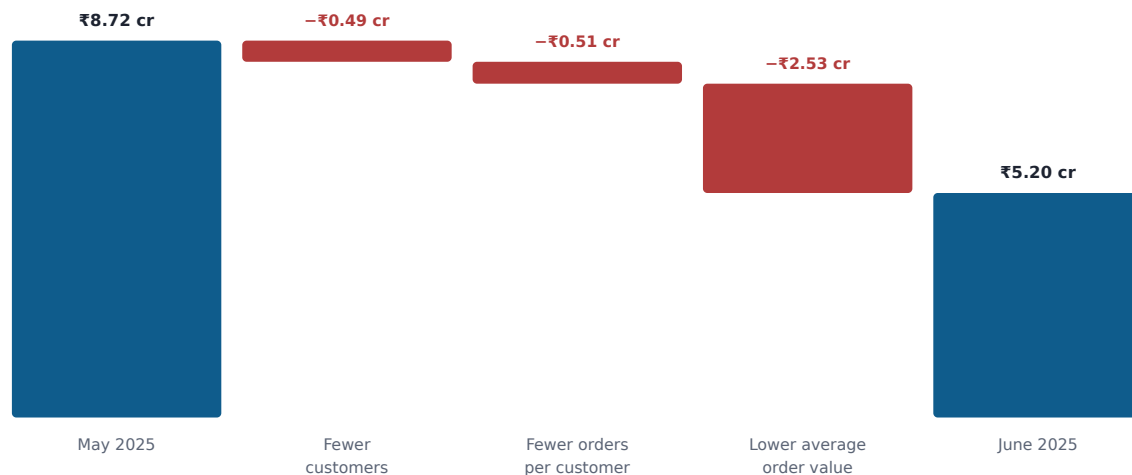


Figure 23.4 — The May → June revenue bridge. Fewer customers and fewer orders each cost a little; a lower average order value cost the most.

This is a **chain-linked (sequential) decomposition**: each effect is measured holding the factors already accounted for at their new value and the rest at the old value, so the three effects sum to the total change exactly, with no leftover residual to explain away. The order you substitute in changes each individual effect slightly, but not the story here.

The finding: of the ₹3.53 crore decline, ₹0.49 crore came from fewer customers, ₹0.51 crore from fewer orders per customer, and **₹2.53 crore — 72% of the fall — from a lower average order value**. Checking the segment mix (Retail, Wholesale, and Hospitality each kept almost exactly their May share of orders) rules out a mix shift: this wasn't "we sold to smaller customers", it was "the same mix of customers placed smaller orders", consistent with the well-established seasonal trough rather than any operational failure.

A general method for diagnosing a KPI move:

1. **Confirm it's real** (Chapter 22): is the change bigger than normal month-to-month noise?
2. **Walk down the tree** one level, and decompose that change the way section 23.11 just did.
3. **Check for a mix effect** (Chapter 22, section 22.6) before believing a within-group story: has the *composition* of what you're measuring changed, or has behaviour genuinely changed?
4. **Compare with a benchmark that isn't your own history**: last year's same month, a similar company, a plan. A seasonal trough looks alarming against last month and unremarkable against last June.
5. **Stop at the level where you have an explanation someone can act on**. "AOV fell because it's the seasonal trough" is an answer; if it weren't seasonal, the next question would be which segment, which branch, which product — one more level down the tree.

23.12 Good metrics, vanity metrics, and gaming

A **vanity metric** goes up reliably and means little: total signups ever, page views, app downloads. It's not dishonest to report; it's just usually the wrong thing to *manage*, because it can rise while the business gets worse.

Vanity metric	Sounds good because	What it hides	Better metric
Total customers ever acquired	Always goes up	Says nothing about who's still active	Active customers this period
Website visits	Big, round, easy to chart	Says nothing about intent or revenue	Leads or conversion rate
Total leads generated	Marketing looks busy	Says nothing about quality	Win rate, or revenue per lead
Features shipped	Feels like progress	Says nothing about whether anyone uses them	Adoption, retention
Gross merchandise value (for a marketplace)	A huge, headline-friendly number	Includes cancelled and returned orders	Net revenue after cancellations

Goodhart's law, informally: *when a measure becomes a target, it stops being a good measure*. The moment a number is tied to a bonus or a promotion, people optimize the number, not necessarily the thing it was meant to represent. This isn't cynicism about people; it's a structural fact about incentives, and Riverstone's own data has an example ready-made:

```
messy = pd.read_csv("../ch14/orders_q4_2025_export.csv", dtype=str, keep_default_na=False)
by_status = messy["status"].str.strip().str.lower().value_counts()
print(by_status)
```

```
status
delivered    21857
pending       1174
shipped       1144
cancelled     1040
dlvd          694
cxl           32
canceled      28
status         6
              1
Name: count, dtype: int64
```

If a branch's bonus depended on "orders delivered", and Chapter 14's Delhi branch had discovered that marking a delayed order "Shipped" rather than leaving it "Pending" moved it out of the bucket anyone was checking, the metric would improve without a single delivery going faster. Nobody has to be dishonest for this to happen; a metric that's easy to move on paper and hard to move in reality *will* get moved on paper, given enough time and enough pressure.

A checklist for spotting a metric at risk of being gamed:

- **Can it be moved without moving the outcome it stands for?** (Marking an order shipped, without shipping it.)

- **Is it a count, when a rate would show the real story?** (Total sales calls made, versus calls that led to a meeting.)
- **Does improving it for one team make it worse somewhere else?** (A "reduce average handling time" target in support can produce rushed, unresolved calls that generate repeat contacts.)
- **Is there a paired metric that would catch the gaming?** (Pair "orders shipped on time" with "orders returned" or "repeat complaints", so gaming one makes the other visible.)

The fix is rarely "stop using metrics." It's to **pair a leading metric with a guardrail metric**, to **audit the definition periodically against the underlying reality** (exactly Chapter 14's reconciliation habit), and to **remember that the number is a proxy, not the goal**.

23.13 Defining a metric so two teams get the same number

Most "our numbers don't match" meetings are not about data quality. They're about two people who never agreed on the definition.

Take "active customer." Reasonable, defensible definitions include: placed an order in the last calendar year; placed an order in the trailing 12 months from today; placed a non-cancelled order; has a non-zero account balance. Each produces a different count from the same database, and none is wrong — they just answer subtly different questions.

A metric definition worth writing down has six parts:

1. **Name.** "Active customer", not "actives".
2. **Formula.** The exact calculation, ideally as pseudocode or SQL, not a sentence that can be read two ways.
3. **Grain and population.** One row per what, over which customers, in Riverstone's case usually excluding the mini-database's Q1 2026 test data.
4. **Time window.** Calendar year, trailing 12 months, or as-of a specific date — and whether it's inclusive of the boundary dates.
5. **What's excluded.** Cancelled orders, internal test accounts, a specific customer segment.
6. **Owner and last-reviewed date.** Someone who can answer questions about it, and evidence it hasn't silently drifted.

```
-- "Active customer, 2025": a documented definition, ready to paste into a data dictionary
-- Owner: Analytics team. Last reviewed: 2026-01-15.
SELECT COUNT(DISTINCT c.customer_id) AS active_customers_2025
FROM customers c
JOIN orders o ON o.customer_id = c.customer_id
WHERE o.status <> 'Cancelled'
      AND o.order_date BETWEEN '2025-01-01' AND '2025-12-31';
```

Write the definition once, put it where both teams can see it (a data dictionary, a semantic layer, a pinned page — Chapter 16's shared semantic model is exactly this idea applied to a BI tool), and change it only with a note explaining why and from when. The alternative — every analyst quietly

making their own reasonable choice — is how a company ends up with three different "revenue" figures in one meeting, all defensible, all disagreeing.

Common mistakes and how to spot them

Mistake	Symptom	Fix
Treating profit and cash as the same thing	"We made money, why can't we pay the supplier?"	Read the cash flow statement; check the cash conversion cycle
Quoting a margin with no rupee figure	"Only 1.8% of revenue" sounds trivial	Give the percentage and the absolute amount together
Reporting EBITDA as if it were cash profit	A leveraged, loss-making company looks "healthy"	State which profit line you mean, and why
Using gross margin to decide whether to take one more order	The decision ignores which costs are actually variable	Contribution margin, not gross margin
Comparing CAC and LTV from different populations	A ratio that looks better or worse than reality	Measure LTV on the customers CAC actually bought
Using the naive 1/churn lifetime for LTV	A years-long assumption nobody would sign off on	Cap the horizon at 3–5 years, and say so
Reporting NPS with no distribution behind it	Two very different customer bases can score the same	Show the distribution, or at least promoters/passives/detractors separately
A vanity metric on the headline slide	"Total signups" always goes up	Pick the metric that can also go down
A KPI tied to a bonus with no guardrail	The number improves; the outcome doesn't	Pair it with a metric that would catch gaming
No agreed definition of a shared metric	Two teams' "revenue" figures disagree	Write the six-part definition and put it where both can see it
Explaining a KPI move without checking mix	A within-group story for what's really a mix shift	Check the mix before believing the story (Chapter 22)
Stopping the diagnosis at the first plausible cause	"Sales fell because of the season" without checking the number	Decompose the tree and quantify each effect
Averaging a ratio across groups of different sizes	The company figure doesn't match any branch	Recompute from the totals, weighted properly (Chapter 21)
Presenting quota attainment as one company average	Hides reps far above and far below target	Show the distribution, not just the mean

In the real world: the profitable company that almost missed payroll

In February 2026, Riverstone's finance controller brought a single sentence to the Monday leadership meeting: *"We are profitable, and next month we may not be able to make payroll."* The room did not immediately believe both halves of that sentence could be true.

Meera had the numbers already, because she'd been asked to explain a smaller, related question the week before: why was the company's cash balance falling even though the P&L showed a healthy year? She walked the room through three facts, in order.

First, the P&L was, on its own terms, good. FY2025 closed at ₹114.7 crore revenue, 27.5% gross margin, and ₹12.1 crore net profit — 10.5% net margin, a respectable result for a distributor. Nobody in the room doubted the year had gone well.

Second, the cash flow statement told a different, narrower story. Operating cash flow was ₹12.9 crore, slightly ahead of profit — normal, even good. But investing activities had used ₹5.2 crore (a new warehouse racking system, approved back in Q2), and financing activities had used a further ₹2.9 crore, mostly the dividend the family had approved based on the profit figure, before anyone had checked what the cash position could actually support. Net cash had risen for the year as a whole, but the *timing* mattered: the dividend and the warehouse capex had both landed in the same eight-week window, right when receivables were also at their seasonal peak.

Third, and this was the number that changed the meeting, the operating cycle itself had quietly lengthened. Meera pulled DSO, DPO, and DIO for the last four quarters instead of the year-end snapshot everyone had been looking at. Days sales outstanding had crept from 36 days at the start of the year to 42 by December, as the sales team — rewarded, reasonably, on revenue booked — had quietly extended payment terms to close bigger Wholesale deals near quarter-end. Inventory days had risen too, as the warehouse stocked up ahead of the festive season. The cash conversion cycle had gone from 51 days to 59 in twelve months, and nobody owned that number or was watching it move.

Two decisions came out of the meeting. The dividend timing moved to align with the low point of the operating cycle rather than the announcement of annual results. And DSO joined the monthly management pack as its own tracked metric, with an owner in Finance and a simple rule: any customer whose payment terms exceeded 45 days needed sign-off from someone other than the rep booking the sale.

Payroll was never actually at risk that particular month — a short-term credit line covered the gap comfortably — but the near-miss was real, and it was caused by exactly the gap this chapter opened with: a profit-only view of a cash problem.

What made the difference:

- **She read the cash flow statement, not just the P&L**, when the P&L said everything was fine.
- **She looked at the trend in DSO and DIO**, not the year-end snapshot, and found a business decision (extended terms to close deals) hiding inside a ratio.

- **She connected an incentive (revenue-based commission) to an outcome (lengthening receivables)** — Goodhart's law, spotted before it became a crisis rather than after.
 - **She turned the finding into an owned, monitored metric**, not just a one-time explanation.
-

Tools

- **Python 3.13 or 3.14** with `pandas` (run here on Python 3.12, `pandas 3.0.2`). Every calculation in this chapter is arithmetic simple enough for a spreadsheet, too.
 - **Excel**: every ratio here is one formula; a KPI tree is a natural fit for a linked set of cells, and Chapter 11's pivot tables handle the segment and monthly breakdowns.
 - **Companion files (`companion/ch23/`):**
 - `build_ch23_files.py` (seed 202301) builds everything below from the real dataset plus documented, invented assumptions.
 - `financials_fy2025.md` : Riverstone's full FY2025 P&L, balance sheet, cash flow statement, and customer/marketing metrics, with every figure labelled real or invented.
 - `monthly_revenue_2025.csv` : the real monthly revenue, orders, customers, and AOV series.
 - `marketing_2025.csv` : monthly marketing spend, leads, new customers, CAC, and ROAS (invented). - Also used: `companion/full/` (leads and lead-stage history, orders), `companion/ch21/delivery_times_2025.csv` , and `companion/ch14/orders_q4_2025_export.csv` .
-

The project: build Riverstone's KPI tree and diagnose a revenue dip

Goal: one page that shows how Riverstone's revenue decomposes, and a second page that diagnoses a real change using that decomposition.

Option A: your own business. Build the tree for a metric you actually own, using your own data.

Option B: Riverstone. Use `companion/full/` , `companion/ch23/` , and `companion/ch21/` .

Steps

1. **Build the top level of the tree:** Revenue = Active customers × Orders per customer × Average order value. Verify the identity reconstructs 2025's actual revenue.
2. **Decompose each branch one level:** active customers into new + retained – churned; AOV into units per order and price/discount.
3. **Attach a real 2025 number to every box**, and name the team that owns each one.
4. **Pick a KPI mentioned elsewhere in this book and place it on the tree** (Chapter 21's on-time delivery rate, Chapter 20's exception rate) — not everything belongs on the revenue tree, and saying why a metric doesn't fit is itself useful.

5. **Diagnose the May → June 2025 dip** using the chain-linked method in section 23.11. Reconstruct the three effects and confirm they sum to the total change.
6. **Check for a mix effect** before concluding: has the segment or branch mix shifted, or is this a within-group change?
7. **Write the one-paragraph diagnosis**, the way section 23.11 does, ending in a sentence someone could act on (or a sentence explaining why no action is needed, if it's a seasonal pattern and nothing more).
8. **Stress-test one metric for gaming**: pick a metric from your tree, and using the section 23.12 checklist, say how it could be moved without moving the underlying outcome, and what guardrail metric would catch it.

What good looks like: the tree's identity reconstructs ₹1,146,641,651 exactly from 4,599 customers × 10.08 orders/customer × ₹24,735.56; the May→June bridge attributes ₹49 lakh to fewer customers, ₹51 lakh to fewer orders per customer, and ₹2.53 crore to lower average order value, summing to the full ₹3.53 crore fall; the segment mix check shows no meaningful shift; and the guardrail exercise names a specific, checkable pairing (for example, "orders shipped" paired with "orders returned within 14 days").

Stretch goals

- Extend the diagnosis to the full year: which month-to-month change was the largest, and does the same three-factor decomposition explain the October festive peak as well as the June trough?
- Build a simple LTV:CAC cohort view: for customers acquired in each quarter of 2025, track their revenue in the following two quarters.
- Draft the six-part definition (section 23.13) for three metrics your own team reports, and check whether a colleague would define any of them differently.

Timed challenge: forty minutes

Use `companion/ch23/financials_fy2025.md`, `monthly_revenue_2025.csv`, and the full dataset. Answers at the end.

- **Level 1:** From the P&L, compute gross margin, EBIT margin, and net margin as percentages.
- **Level 2:** From the balance sheet, compute working capital, the current ratio, and debt-to-equity.
- **Level 3:** Compute the cash conversion cycle from the given DIO, DSO, and DPO.
- **Level 4:** From the cash flow statement, explain in one sentence why cash rose by less than profit after tax.
- **Level 5:** Using `monthly_revenue_2025.csv`, find the two consecutive months with the largest percentage fall in revenue, and compute it.
- **Level 6:** Decompose that fall into customer, frequency, and AOV effects using the chain-linked method.
- **Level 7:** State Riverstone's retention rate, churn rate, and 5-year-capped LTV, and compare LTV:CAC against the 3:1 rule of thumb.

- **Bonus:** Using the real leads data, compute the funnel conversion rate from "Contacted" to "Won" (not from "New").
-

You've got it when...

- ☐ You can explain the operating cycle and why a profitable company can run out of cash.
 - ☐ You can read a P&L, a balance sheet, and a cash flow statement and say what each answers.
 - ☐ You know the difference between gross, EBIT, and net margin, and between EBIT and EBITDA.
 - ☐ You can compute and interpret the core metrics of sales, marketing, finance, operations, and customer teams.
 - ☐ You never quote CAC or LTV without saying which population and which time horizon.
 - ☐ You can build a KPI tree that reconstructs a real total exactly, with each branch owned by a team.
 - ☐ You diagnose a metric change by decomposing it, not by guessing, and you check for a mix effect first.
 - ☐ You can name a vanity metric and its better replacement.
 - ☐ You can predict how a metric would be gamed if a bonus depended on it, and propose a guardrail.
 - ☐ You write metric definitions with a formula, a population, a time window, exclusions, and an owner.
-

Recap

- **The operating cycle** turns cash into stock into sales into cash again. Riverstone's cash conversion cycle is 59 days (55 DIO + 42 DSO – 38 DPO); profit and cash are not the same thing, and the gap between them is exactly this cycle.
- **The P&L** shows revenue (₹114.7 cr) minus COGS gives gross profit (27.5% margin); minus operating expenses gives EBIT (15.0%); minus interest and tax gives net profit (10.5%).
- **The balance sheet** balances by definition (assets = liabilities + equity) and is where receivables and inventory – both analyst territory – live.
- **The cash flow statement** reconciles profit with the actual change in cash, across operating, investing, and financing activities.
- **Sales metrics** (win rate, sales cycle, AOV, quota attainment) describe the path from lead to revenue; Riverstone's real pipeline shows a 60% win rate and a 25.5-day cycle.
- **Marketing metrics** (CAC, ROAS, the funnel) always depend on an attribution model – state it.
- **Finance metrics** add contribution margin, ROA, and ROE to the vocabulary from the statements.
- **Operations metrics** (OTIF, inventory turns, fill rate) are the same "profile the distribution" skill from Chapter 21, in industry language.
- **Customer metrics** (retention, churn, LTV, LTV:CAC, NPS) need a capped time horizon and a matched population; Riverstone's 93.3% retention and 26:1 LTV:CAC are both explainable, not

automatically suspicious.

- **A KPI tree** decomposes a headline number into an exact identity, each branch owned by a team; diagnosing a change means walking the tree and quantifying each effect, checking for a mix shift before trusting a within-group story.
- **Vanity metrics** always rise and mean little; a metric tied to an incentive will be gamed unless paired with a guardrail.
- **A metric definition** needs a name, formula, grain, time window, exclusions, and owner — written down once, not reinvented by every analyst who touches it.

Practice exercises

Use `companion/ch23/`, `companion/full/`, and `companion/ch21/delivery_times_2025.csv`.

Warm-up

1. A friend says "the business made a profit this year, so it must have more cash than last year." Explain why that isn't guaranteed, using the three financial statements.
2. What's the difference between gross margin and net margin? Give a one-sentence business reason a company might have high gross margin and low net margin.
3. Riverstone's DIO is 55 days, DSO is 42, DPO is 38. Compute the cash conversion cycle, and explain in one sentence what a *negative* CCC would mean for a company.
4. Name the three sections of a cash flow statement and one Riverstone example of an item in each.
5. Why is "total leads generated" a vanity metric for a sales team, and what would you track instead?
6. Write a six-part definition for "active customer" as Chapter 23 would use it.

Core

7. Compute Riverstone's EBITDA (EBIT + depreciation) and its margin. Why might a lender care about EBITDA more than net profit?
8. Compute the current ratio and debt-to-equity from the balance sheet. Would you describe Riverstone as conservatively or aggressively financed?
9. Using the real leads data, compute the conversion rate at each stage of the funnel (New→Contacted, Contacted→Quoted, Quoted→Won). Which stage loses the most leads?
10. Compute Riverstone's blended CAC and average ROAS from `marketing_2025.csv`. In which month was ROAS highest, and does that month's lead volume explain it?
11. Compute the naive (1/churn) and 5-year-capped LTV, and the resulting LTV:CAC for each. Which would you present to a board, and why?
12. Verify the KPI tree identity for 2025: active customers × orders per customer × AOV. Confirm it reconstructs the real revenue figure.
13. Find the two consecutive months in 2025 with the largest *percentage* fall in revenue, and decompose it into customer, frequency, and AOV effects.

14. For the change in exercise 13, check whether the segment mix shifted meaningfully between the two months.
15. Compute inventory turns ($\text{COGS} \div \text{average inventory}$, using the year-end inventory figure as a simple proxy for the average). What would doubling inventory turns imply about DIO?
16. Using Chapter 21's delivery data, compute OTIF by branch and rank the branches. Propose one guardrail metric that would catch a branch gaming OTIF by relabelling status.
17. Using the messy Q4 export from Chapter 14, count how many rows are some variant of "Shipped". If a branch's bonus depended on minimizing "Pending" orders, what behavior would you watch for?
18. Write the SQL (or pandas) for a second, differently-scoped definition of "active customer" (for example, trailing 12 months from 31 December 2025 instead of calendar year), and show that it gives a different count.
19. Compute return on equity and return on assets. Which is higher, and what does the gap tell you about how the business is financed?
20. Draft a one-paragraph answer, in the style of section 23.11, diagnosing October 2025's revenue peak instead of the June trough.

Stretch

21. Build the full second level of the KPI tree (new/retained/churned customers; units per order and price/discount for AOV) with real 2025 numbers in every box.
22. Compute a cohort LTV: for customers whose first order was in Q1 2025, what was their total revenue through Q4 2025? Compare it with the chapter's blended LTV estimate.
23. Propose Riverstone's North Star metric, with a one-paragraph justification, and explain what it would and wouldn't tell leadership that revenue doesn't.
24. Using `marketing_2025.csv`, compute the payback period ($\text{CAC} \div \text{monthly gross profit per new customer}$) and say whether Riverstone's marketing spend is cash-efficient.

Think about it

25. Riverstone's board asks why the company should invest more in marketing when it's already profitable. Using this chapter's numbers, make the case for and against.
26. A regional sales manager's bonus depends entirely on quarterly revenue booked. Predict two ways this could distort the numbers you'd see in the data, and propose a guardrail for each.
27. Two departments each report "customer retention" and get different numbers from the same underlying data. What's your first question, and what would you do about it going forward?

Key terms

operating cycle · DIO · DSO · DPO · cash conversion cycle · profit & loss (P&L) · income statement · revenue · cost of goods sold (COGS) · gross profit / margin · operating expenses (opex) · EBIT · EBITDA · net profit (PAT) · net margin · accrual accounting · balance sheet · assets · liabilities · equity · current

assets / liabilities · working capital · current ratio · debt-to-equity · cash flow statement · CFO / CFI / CFF · fixed cost · variable cost · contribution margin · pipeline · win rate · sales cycle · quota attainment · funnel · CAC · ROAS · attribution (first-touch, last-touch, multi-touch) · payback period · return on assets (ROA) · return on equity (ROE) · OTIF · inventory turns · fill rate · OEE · retention rate · churn rate · customer lifetime value (LTV) · LTV:CAC · Net Promoter Score (NPS) · KPI tree · North Star metric · chain-linked (sequential) decomposition · mix effect · vanity metric · Goodhart's law · guardrail metric · metric definition · data dictionary

(All terms are defined in the Glossary, Appendix A.)

Where this leads

- **Chapter 24, Requirements, Storytelling & Stakeholders:** turning a diagnosis like section 23.11's into a memo and a three-slide story someone will act on.
- **Chapter 25, The Business Analyst Track:** process mapping and requirements gathering, which is how a KPI tree gets built collaboratively rather than assumed.
- **Chapter 16, Business Intelligence with Power BI:** a KPI tree is naturally a dashboard, with the shared semantic model as the enforcement mechanism for section 23.13's metric definitions.
- **Chapter 22, Statistics Without Fooling Yourself:** confirming a KPI move is real before diagnosing it, and checking for confounders and mix effects.
- **Part IV:** LTV modelling, churn prediction, and marketing mix modelling all extend the ideas in sections 23.6 and 23.9 with statistical machinery.
- **Interview preparation:** the Business Analyst bank (Chapter 76) and the Case Study & Guesstimate bank (Chapter 74) both lean heavily on exactly this chapter — "walk me through how you'd diagnose a revenue drop" is one of the most common analyst interview questions there is.

Answers to practice exercises

(In the finished book these move to Appendix G.)

1. Profit is booked when a sale is made, not when it's paid for; a company can be profitable while its cash is tied up in growing receivables and inventory (the operating cycle), or while it spends heavily on capex or dividends. Only the cash flow statement shows what actually happened to the bank balance.
2. Gross margin is revenue minus the cost of the product itself; net margin is what's left after every other cost, including overhead, marketing, interest, and tax. A software company can have very high gross margin (little "cost of goods") and low net margin if it spends heavily on sales, marketing, and R&D.
3. $CCC = 55 + 42 - 38 = 59$ **days**. A negative CCC (common in retail and e-commerce) means the company collects from customers before it has to pay its own suppliers — effectively financing its inventory with someone else's money.

4. Operations: profit adjusted for depreciation and working-capital changes. Investing: the ₹5.16 crore capital expenditure (warehouse racking, in the story). Financing: the ₹4.23 crore dividend, debt drawn, and loan repayment.

5. It rises whenever marketing runs any campaign, regardless of whether those leads ever buy anything or fit the business. Track win rate, or revenue per lead, which can go down even while lead count goes up.

6. Name: Active customer, 2025. Formula: `COUNT(DISTINCT customer_id) from orders with status <> 'Cancelled'`. Grain/population: Riverstone's full customer base, excluding the mini-database's test data. Time window: calendar year 2025, inclusive of both boundary dates. Exclusions: cancelled orders. Owner: Analytics team, last reviewed [date].

7. EBITDA = EBIT + depreciation = ₹17.15 cr + ₹1.61 cr = **₹18.76 crore**, margin **16.4%**. A lender cares about EBITDA because it approximates cash available to service debt, independent of how assets happen to be depreciated or financed — closer to "can this business make its loan payments" than net profit is.

8. Current ratio 1.58, debt-to-equity 0.91. A current ratio comfortably above 1 and debt roughly equal to (a bit below) equity is a moderately, not aggressively, financed balance sheet — enough leverage to grow, enough cushion to survive a bad quarter.

9. New→Contacted: 22/43 ≈ 51%. Contacted→Quoted: 14/22 ≈ 64%. Quoted→Won: 6/14 ≈ 43% (with 4 of 14 Lost and the rest still open). The New→Contacted stage loses the most leads in absolute terms (21 of 43 never get contacted).

10. Blended CAC ≈ ₹8,194; average ROAS ≈ 5.6x. The highest ROAS month is March (7.37x), which had strong lead volume (470) and an above-average conversion rate that month — not simply the month with the most leads (October had more leads but a lower ROAS).

11. Naive LTV (14.8-year lifetime): ₹6.42 lakh, LTV:CAC ≈ 78:1. 5-year-capped: ₹2.17 lakh, LTV:CAC ≈ 26:1. Present the capped figure to a board: it rests on an assumption (5 years of continued purchasing) that's far easier to defend than fourteen years of extrapolation from one year of churn data.

12. $4,599 \times (46,356/4,599) \times ₹24,735.56 = 4,599 \times 10.08 \times 24,735.56 \approx$ **₹1,146,641,651**, matching the real 2025 revenue exactly (the identity is definitional, so it always will).

13. The largest percentage fall is **November → December** (−44.1%, from ₹15.60 crore to ₹8.73 crore), narrowly ahead of May→June (−40.4%). Decomposed: customer effect ≈ −₹1.26 crore, frequency effect ≈ −₹1.06 crore, AOV effect ≈ −₹4.55 crore, summing to the full −₹6.87 crore fall — AOV is again the dominant driver, at 66% of the decline.

14. For November→December, segment order-share is nearly unchanged (Hospitality 27.5%→26.8%, Retail 50.5%→52.0%, Wholesale 22.0%→21.1%), so the AOV decline is again a broad, within-segment effect (the post-festive normalisation) rather than a shift toward smaller-spending segments.

15. $\text{COGS} \div \text{inventory} = 831,803,000 \div 125,340,178 \approx$ **6.6 times a year**. Doubling turns to about 13.2 would imply DIO falling from 55 days to roughly 27.5 days — much less cash tied up in stock, but a higher risk of stockouts unless replenishment also gets faster.

16. OTIF by branch: Mumbai HO 90.8%, Delhi 80.0%, Bengaluru 78.0%, Kolkata 68.1%. A guardrail: pair "on-time rate" with "customer-reported late deliveries" or "orders relabelled after the promised date", so relabelling an order's status without it actually arriving would show up as a gap between the two.

17. 1,144 rows are some spelling of "Shipped" (plus 21,857 Delivered, 1,174 Pending, and cancelled variants). If a bonus depended on minimizing Pending orders, watch for orders moved to "Shipped" status without a corresponding dispatch record or tracking update — exactly the gap a paired metric (dispatch confirmations, or actual delivery dates) would expose.

18. For example, `WHERE order_date BETWEEN '2025-01-01' AND '2025-12-31'` (calendar year) versus `WHERE order_date > '2024-12-31'::date - INTERVAL '365 days'` measured from 31 December 2025 (trailing twelve months) will generally differ by a handful of customers whose activity falls just outside one window and inside the other — small in percentage terms, but enough to produce "different" numbers in a meeting if nobody states which definition they used.

19. $ROE = 120,891,502 / 378,225,976 \approx 32.0\%$; $ROA = 120,891,502 / 722,819,016 \approx 16.7\%$. ROE well above ROA indicates the company is using debt (leverage) to boost the return to equity holders above what the assets alone generate.

20. October 2025 revenue rose to ₹18.06 crore from September's ₹11.17 crore, a ₹6.89 crore increase. Decomposing the same way as the June dip: more active customers, more orders per customer, and a higher average order value all contributed, consistent with the well-documented festive-season peak (Chapters 11 and 15) rather than any one-off event. The segment mix shows Wholesale's share of orders rising slightly, consistent with bulk festive restocking.

21. New customers (767) + retained (3,832) – churned (277) reconciles to the 2025 active count of 4,599 minus the double-count adjustment (new + retained = 4,599 exactly, since every 2025 customer is either new or retained by definition; churned describes 2024's base, not 2025's). Units per order and price/discount multiply to the ₹24,735.56 AOV.

22. A cohort LTV (actual revenue from a specific quarter's new customers, tracked forward) is always more trustworthy than a blended, formula-based estimate, because it has no assumptions baked in — only the risk that the cohort isn't yet finished generating revenue. Expect it to run somewhat below the 5-year-capped estimate, since it only covers nine months rather than five years.

23. A reasonable choice is **active customers**, because it sits upstream of both revenue (via frequency and order size) and retention, and it's a number every function can influence. It would tell leadership whether the customer base is growing or shrinking *before* that shows up in a revenue number that can be temporarily flattered by a few large orders; it would not, on its own, say anything about profitability.

24. Payback period = $CAC \div (\text{monthly gross profit per new customer})$. With average first-year revenue of ₹157,720 and 27.5% margin, monthly gross profit per new customer $\approx ₹3,615$, giving a payback of roughly 2.3 months on a CAC of ₹8,194 — a fast payback that supports the case for spending more, not less, on acquisition.

25. For: LTV:CAC of 26:1 is far above the 3:1 rule of thumb, payback is under three months, and the company is already profitable with cash to deploy — textbook conditions for investing more in acquisition while it's working. Against: the 26:1 ratio rests on assumptions (the 5-year horizon, the

attribution model) that haven't been stress-tested at higher spend, and past a point, additional marketing spend usually raises CAC (the easiest customers are acquired first) faster than revenue grows to match.

26. (a) Revenue could be pulled forward by discounting or extending payment terms near quarter-end (exactly the DSO story in this chapter's real-world example) — a guardrail would track DSO and discount rates by rep alongside revenue. (b) Deals could be booked before they're actually won, or split to appear as multiple smaller wins — a guardrail would track the rate of cancelled or reversed orders shortly after being recorded.

27. First question: "what's your exact definition — formula, time window, and what's excluded?" Very often the discrepancy is entirely explained by one team using calendar year and the other trailing twelve months, or one excluding cancelled orders and the other not. Going forward: write the six-part definition once, in a shared location, with an owner, so the next disagreement takes five minutes instead of a meeting.

Timed challenge answers. Level 1: gross margin 27.5%, EBIT margin 15.0%, net margin 10.5%. Level 2: working capital ₹12.60 crore, current ratio 1.58, debt-to-equity 0.91. Level 3: CCC = $55 + 42 - 38 = 59$ days. Level 4: cash rose by less than PAT because ₹5.16 crore was spent on capex and ₹4.23 crore paid out as dividend, both financing/investing outflows that don't appear in profit. Level 5: November→December, a fall from ₹15.60 crore to ₹8.73 crore, about 44.1% (the largest percentage fall of the year; May→June at -40.4% is a close second). Level 6: customer effect $\approx$ -₹1.26 crore, frequency effect $\approx$ -₹1.06 crore, AOV effect $\approx$ -₹4.55 crore, summing to the full -₹6.87 crore fall. Level 7: retention 93.3%, churn 6.7%, 5-year-capped LTV $\approx$ ₹2.17 lakh, LTV:CAC $\approx$ 26:1 — well above the 3:1 rule of thumb. Bonus: Contacted→Won using the funnel (6 Won of 22 that reached Contacted) $\approx$ 27%.